



CMA 6134 - COMPUTATIONAL METHOD

FACILITY OF COMPUTER AND INFORMATICS

TRIMESTER : 2510

Coding Assignment Report

Lecture: TC2L

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NO	NAME	STUDENT ID
1.	Teh Shin Rou	243UC2467K
2.	Bianca Lau Ying Xuan	242UC2426R
3.	Nyiam Zi Qin	243UC2466T
4.	Wong Yong Kit	242UC244V7

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1.0 Introduction

This report describes a simulation of a petrol station with four pumps and two lanes, aiming to study how vehicles are served during both peak and non peak hours. The simulation models random vehicle arrivals, types of vehicles(motorbike, car, lorry) fuel types and the time taken for refueling. It helps to understand how long vehicles wait, how long they stay in the system, and how efficiently each pump is used.

The main idea is to see how real-life conditions like traffic flow and vehicle types affect service time and waiting queues. By stimulating different scenarios, we can measure things like average waiting time, probability of waiting, and pump workload. This can give useful insights into how petrol stations operate and where improvements can be made.

2.0 Methodology

2.1 Arrival Pattern

```
% --- Start Simulation Loop ---  
for i = 1:n  
    rInter = rand; % Random number for inter-arrival time  
    if p == 1  
        interArrival = floor(rand * 2) + 1;  
    else  
        interArrival = floor(rand * 4) + 3;  
    end  
    arrivalTime = lastArrival + interArrival;
```

Vehicle arrival times are stimulated using random numbers to represent natural variations in traffic flow. During peak hours($p=1$), vehicles arrive every 1 to 2 minutes. During non-peak hours($p=0$), interarrival times are longer which arrive every 3 to 6 minutes. This logic helps stimulate congestion during busy times and lighter flow during off-hours.

2.2 Vehicle Categories and Refueling Time

```
rRefuel = rand; % Random number for refueling (vehicle type)
if rRefuel < 0.3
    vType = 'Motorbike'; fuelOptions = [1,2];
    qty = floor(rand * 4) + 3; refuelTime = floor(rand * 3) + 2;
elseif rRefuel < 0.9
    vType = 'Car'; fuelOptions = [1,2];
    qty = floor(rand * 31) + 20; refuelTime = floor(rand * 5) + 4;
else
    vType = 'Lorry'; fuelOptions = [3];
    qty = floor(rand * 41) + 60; refuelTime = floor(rand * 5) + 6;
end
```

Each vehicle is then randomly categorized as a motorbike, car or lorry. The distribution is based on probabilities, with motorbikes making up about 30%, cars 60% and lorries 10% of the traffic. Motorbikes typically refuel for 2 to 4 minutes and use 3 to 6 litres of fuel. Cars require 4 to 8 minutes and consume 20 to 50 litres, while lorries take the longest- 6 to 10 minutes and use 60 to 100 litres of fuel. The type of vehicle influences the amount of fuel used and the time required for refueling which directly affects waiting time and pump availability.

2.3 Fuel Type

```
petrolIndex = fuelOptions(floor(rand * length(fuelOptions)) + 1);
petrol = petrolTypes{petrolIndex};
pricePerLitre = prices(petrolIndex);
totalPrice = qty * pricePerLitre;
```

Once the vehicle type is selected, the simulation assigns a compatible fuel type. Motorbikes and cars can choose between Primax95 and Primax97, while lorries use Dynamic Diesel only. This is randomly decided for eligible vehicles. Each fuel type has a fixed price per litre, and the total cost for each vehicle is calculated by multiplying the quantity of fuel used with the corresponding price. So, the stimulation not only tracks time-based performance but also the spending per customer.

2.4 Pump and Lane

```
% Lane 1: Pumps 1 and 2
% Lane 2: Pumps 3 and 4

% Find earliest available pump in Lane 1
earliestLane1 = min(pumps(1), pumps(2));

% Find earliest available pump in Lane 2
earliestLane2 = min(pumps(3), pumps(4));

% Introduce a random tie-breaker if both lanes are equally available
if earliestLane1 < earliestLane2
    laneUsed(i) = 1;
    if pumps(1) <= pumps(2)
        pumpID = 1;
    else
        pumpID = 2;
    end
elseif earliestLane2 < earliestLane1
    laneUsed(i) = 2;
    if pumps(3) <= pumps(4)
        pumpID = 3;
    else
        pumpID = 4;
    end
else %earliestLane1 == earliestLane2
    %Randomly choose between Lane 1 and Lane 2
    if rand() < 0.5 %50% chance to go to Lane 1
        laneUsed(i) = 1;
        if pumps(1) <= pumps(2)
            pumpID = 1;
        else
            pumpID = 2;
        end
    else %Or else go to Lane 2
        laneUsed(i) = 2;
        if pumps(3) <= pumps(4)
            pumpID = 3;
        else
            pumpID = 4;
        end
    end
end
```

Pump assignment is handled by comparing the availability of pumps in two lanes. Lane 1 contains Pump 1 and Pump 2, while Lane 2 contains Pump 3 and Pump 4. For each arriving vehicle, the program checks which lane has the earliest available pump. If one lane is available soon, the vehicle is assigned to that lane. If both lanes are available at the same time, a random decision is made to fairly distribute vehicles between them. Within the selected lane, the pump with the shortest queue or idle time is chosen.

2.5 Timing and Queue Calculation

```
%Calculate timings
startTime = max(arrivalTime, pumps(pumpID));
endTime   = startTime + refuelTime;
waitTime  = startTime - arrivalTime;
timeSpent = endTime - arrivalTime;
```

The simulation calculates key time values for each vehicle. These include the time they arrive, the time they begin refueling, the time refueling ends, the amount of time they have to wait, and their total time in the system. These data points are stored and used to generate average waiting time, average time in the system, the probability that a vehicle had to wait and the average service time per pump.

3.0 Stimulation Design

This stimulation was created using MATLAB to model how vehicles arrive and get served at a petrol station with four pumps. The main purpose is to understand how different types of vehicles are handled during peak and non-peak hours, and to track their waiting times, fuel usage, and how busy each pump gets.

The code starts by setting up all the needed variables and arrays to store details for each vehicle, like arrival time, how long they wait, what fuel they use, and which pump they go to. The four pumps are split into two lanes-Lane 1 has Pump 1 and Pump 2, and Lane 2 has Pump 3 and Pump 4. Each pump is tracked to know when it's free to serve the next vehicle.

There are three key tables printed at the beginning: interarrival time, fuel type, and refueling time. These help define the random behavior in the simulation. For interarrival time, if it's a peak hour, vehicles arrive every 1 or 2 minutes. In non-peak times, the gap between vehicles can be 3 to 6 minutes. The arrival time is calculated by adding the interval to the last arrival.

The fuel type is chosen randomly based on set probabilities: Primax95 (45%), Primax97 (45%) and Dynamic Diesel (10%). Motorbikes and cars can use either of the Primax fuels, but lorries always use diesel. Each type of fuel has a fixed price, and this is used to calculate how much the driver has to pay.

Vehicles are also randomly assigned a type: motorbike, car or lorry. Each type has a range for how much fuel they use and how long they take a refuel. Motorbikes are fastest (2-4 minutes), cars take longer(4-8 minutes) and lorries take the most time(6-10 minutes).

When assigning pumps, the program checks which of the two lanes has a pump that will be free the earliest. If both are equally available, it randomly picks one. Once a vehicle is assigned a pump, it either starts refueling right away or waits until the pump is free. The actual start time is based on the later of the vehicle's arrival or when the pump becomes available.

All information of each vehicle arrives, starts refueling, ends and leaves is stored in an event log. At the end, the simulation prints all the vehicle details in a clear table, along with results like average waiting time, chance of waiting and how long vehicles stayed in the system. It also shows how busy each pump was on average.

4.0 Results

4.1 Simulation Parameters

- **Number of Vehicles (n):** The simulation was conducted for 50 vehicles for both non-peak and peak hour scenarios.

Inter-Arrival Time Distribution:

- **Non-Peak Hour (p=0):** Vehicles arrive with a wider, longer inter-arrival time distribution with inter-arrival times of **3, 4, 5, or 6 minutes**, each with a 25% probability. This means vehicles are spaced out more, leading to fewer simultaneous arrivals.
- **Peak Hour (p=1):** Vehicles arrive with a much narrower, shorter inter-arrival time distribution with inter-arrival times of **1 or 2 minutes**, each with a 50% probability. This signifies a higher frequency of vehicle arrivals.

Impact on Congestion and Waiting:

- **Non-Peak Hour:** Due to the longer inter-arrival times, pumps are more likely to be available when a new vehicle arrives. This results in:
 - **Lower average waiting times.**
 - **Lower probability of waiting.**
 - **Lower average time spent in the system.**
 - **Potentially lower pump utilization.**
- **Peak Hour:** The rapid succession of vehicle arrivals puts significant pressure on the limited number of pumps. This leads to:
 - **Significantly higher average waiting times.**
 - **Much higher probability of waiting.**
 - **Higher average time spent in the system.**
 - **Higher pump utilization.**

4.2 Probability Distributions Used

Table 4.2.1: Inter-arrival Time Distribution

Inter-arrival time	Probability	CDF	Random number range
Non-Peak Hour ($p=0$)			
3 min	0.25	0.25	0.00 ~ 0.25
4 min	0.25	0.50	0.25 ~ 0.50
5 min	0.25	0.75	0.50 ~ 0.75
6 min	0.25	1.00	0.75 ~ 1.00
Peak Hour ($p=1$)			
1 min	0.50	0.50	0.00 ~ 0.50
2 min	0.50	1.00	0.50 ~ 1.00

Table 4.2.2: Type of Petrol Distribution

Type of petrol	Probability	CDF	Random number range	Price/Litre (RM)
Primax95	0.45	0.45	0.00 ~ 0.45	2.05
Primax97	0.45	0.90	0.45 ~ 0.90	2.50
Dynamic Diesel	0.10	1.00	0.90 ~ 1.00	2.15

Table 4.2.3: Refueling Time Distribution

Vehicle	Refueling Time Range	Probability	CDF	Random number range
Motorbike	2~4 min	0.30	0.30	0.00 ~ 0.30
Car	4~8 min	0.60	0.90	0.30 ~ 0.90
Lorry	6~10 min	0.10	1.00	0.90 ~ 1.00

4.3 Summary of Simulation Outcomes

Table 4.3.1: Final Simulation Results for Non-Peak Hour (p=0)

Metric	Value (Minutes)	Value (Min:Sec)
Average Waiting Time	0.00	0 min 0 sec
Average Time in System	5.18	5 min 11 sec
Probability of Waiting	0.00	N/A

Table 4.3.2: Average Service Time Per Pump (Non-Peak Hour, p=0)

Pump ID	Average Service Time (Minutes)	Average Service Time (Min:Sec)
1	5.46	5 min 28 sec
2	4.92	4 min 55 sec
3	5.01	5 min 5 sec
4	5.25	5 min 15 sec

Table 4.3.3: Final Simulation Results for Peak Hour (p=1)

Metric	Value (Minutes)	Value (Min:Sec)
Average Waiting Time	0.96	0 min 58 sec
Average Time in System	6.62	6 min 37 sec
Probability of Waiting	0.44	N/A

Table 4.3.4: Average Service Time Per Pump (Peak Hour, p=1)

Pump ID	Average Service Time (Minutes)	Average Service Time (Min:Sec)
1	5.92	5 min 55 sec
2	5.54	5 min 32 sec
3	5.92	5 min 55 sec
4	5.31	5 min 18 sec

4.4 Simulation Table

Detailed simulation tables were also produced as the tables shown below. These tables were instrumental in validating the simulation logic and calculating aggregate statistics.

4.1 Peak Hour Simulation Table

Vehicle number	Type of vehicle	Type of petrol	Quantity (liter)	Total price (RM)
1	Car	Primax97	21	52.50
2	Car	Primax97	26	65.00
3	Lorry	Dynamic Diesel	65	139.75
4	Car	Primax95	31	63.55
5	Car	Primax97	45	112.50
6	Motorbike	Primax97	3	7.50
7	Car	Primax95	33	67.65
8	Car	Primax95	35	71.75
9	Car	Primax95	29	59.45
10	Car	Primax97	39	97.50

Random number for inter-arrival time	Inter-arrival time	Arrival time	Line number	Random number for refueling
0.14	2	2	1	0.83
0.28	1	3	1	0.37
0.48	2	5	2	1.00
0.48	1	6	2	0.57
0.84	2	8	1	0.70
0.30	2	10	1	0.28
0.29	2	12	2	0.85
0.80	1	13	1	0.83
0.08	1	14	2	0.39
0.30	2	16	1	0.61

4.4.2 Peak Hour Simulation Table

Vehicle number	Pump 1			Pump 2		
	Refueling time	Time begins	Time ends	Refueling time	Time begins	Time ends
1	4	2	6	-	-	-
2	-	-	-	7	3	10
3	-	-	-	-	-	-
4	-	-	-	-	-	-
5	8	8	16	-	-	-
6	-	-	-	4	10	14
7	-	-	-	-	-	-
8	-	-	-	5	14	19
9	-	-	-	-	-	-
10	7	16	23	-	-	-

Pump 3			Pump 4			Waiting time	Time spent
Refueling time	Time begins	Time ends	Refueling time	Time begins	Time ends		
-	-	-	-	-	-	0	4
-	-	-	-	-	-	0	7
6	5	11	-	-	-	0	6
-	-	-	8	6	14	0	8
-	-	-	-	-	-	0	8
-	-	-	-	-	-	0	4
7	12	19	-	-	-	0	7
-	-	-	-	-	-	1	6
-	-	-	8	14	22	0	8
-	-	-	-	-	-	0	7

4.4.3 Non-Peak Hour Simulation Table

Simulation Table

Vehicle number	Type of vehicle	Type of petrol	Quantity (liter)	Total price (RM)
1	Motorbike	Primax97	6	15.00
2	Car	Primax95	34	69.70
3	Car	Primax97	45	112.50
4	Motorbike	Primax97	6	15.00
5	Car	Primax95	50	102.50
6	Car	Primax97	44	110.00
7	Lorry	Dynamic Diesel	62	133.30
8	Lorry	Dynamic Diesel	77	165.55
9	Motorbike	Primax97	4	10.00
10	Motorbike	Primax97	3	7.50

Random number for inter-arrival time	Inter-arrival time	Arrival time	Line number	Random number for refueling
0.45	5	5	1	0.21
0.60	4	9	1	0.61
0.23	5	14	2	0.51
0.11	5	19	2	0.11
0.58	3	22	1	0.77
0.27	4	26	1	0.62
0.28	3	29	2	0.96
0.98	6	35	2	0.95
0.24	4	39	1	0.04
0.22	4	43	1	0.25

4.4.4 Non-Peak Hour Simulation Table

Vehicle number	Pump 1			Pump 2		
	Refueling time	Time begins	Time ends	Refueling time	Time begins	Time ends
1	2	5	7	-	-	-
2	-	-	-	4	9	13
3	-	-	-	-	-	-
4	-	-	-	-	-	-
5	8	22	30	-	-	-
6	-	-	-	5	26	31
7	-	-	-	-	-	-
8	-	-	-	-	-	-
9	4	39	43	-	-	-
10	-	-	-	2	43	45

Pump 3			Pump 4			Waiting time	Time spent
Refueling time	Time begins	Time ends	Refueling time	Time begins	Time ends		
-	-	-	-	-	-	0	2
-	-	-	-	-	-	0	4
7	14	21	-	-	-	0	7
-	-	-	3	19	22	0	3
-	-	-	-	-	-	0	8
-	-	-	-	-	-	0	5
8	29	37	-	-	-	0	8
-	-	-	10	35	45	0	10
-	-	-	-	-	-	0	4
-	-	-	-	-	-	0	2

4.5 Event Log

A comprehensive chronological event log was generated as shown in the images below. This log provides a granular view of the simulation flow.

4.5.1 Peak Hour Event Log

Vehicle 1 arrived at minute 1 and began refueling with Primax95 at Pump Island 3.
Vehicle 2 arrived at minute 2 and began refueling with Primax95 at Pump Island 1.
Vehicle 3 arrived at minute 4 and began refueling with Primax95 at Pump Island 4.
Vehicle 4 arrived at minute 5 and began refueling with Primax95 at Pump Island 2.
Vehicle 1 finished refueling and departed at minute 6.
Vehicle 5 arrived at minute 7 and began refueling with Primax95 at Pump Island 3.
Vehicle 6 arrived at minute 8 and began refueling with Primax95 at Pump Island 1.
Vehicle 2 finished refueling and departed at minute 9.
Vehicle 4 finished refueling and departed at minute 9.
Vehicle 3 finished refueling and departed at minute 10.
Vehicle 7 arrived at minute 10 and began refueling with Primax95 at Pump Island 2.
Vehicle 5 finished refueling and departed at minute 11.
Vehicle 8 arrived at minute 12 and began refueling with Primax97 at Pump Island 4.
Vehicle 7 finished refueling and departed at minute 14.
Vehicle 6 finished refueling and departed at minute 14.
Vehicle 9 arrived at minute 14 and began refueling with Primax97 at Pump Island 3.
Vehicle 10 arrived at minute 16 and began refueling with Primax95 at Pump Island 1

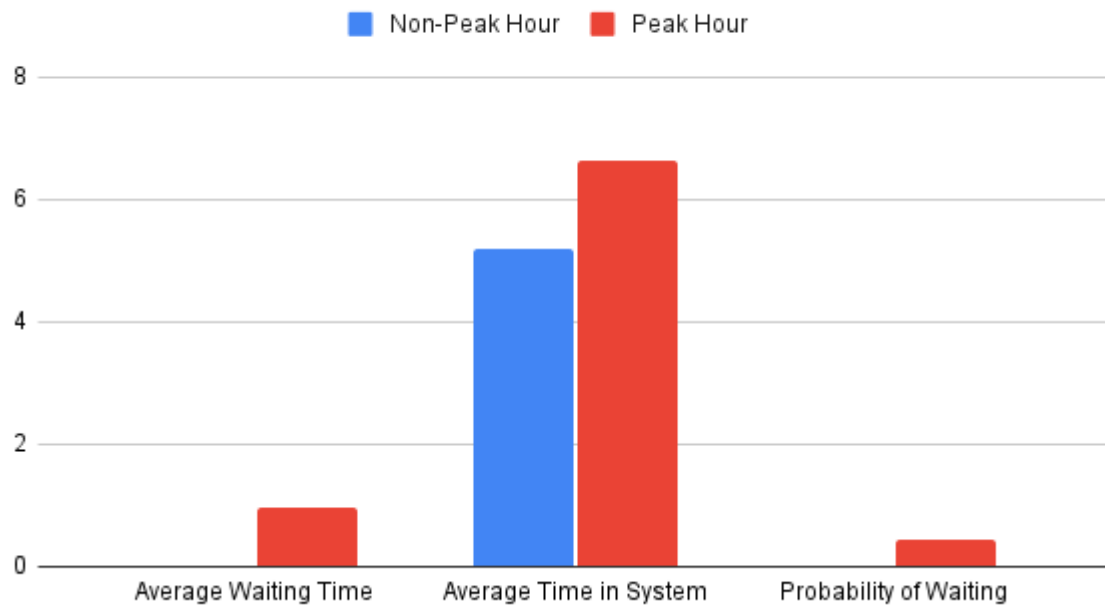
4.5.1 Peak Hour Event Log

Vehicle 1 arrived at minute 5 and began refueling with Primax95 at Pump Island 3.
Vehicle 1 finished refueling and departed at minute 9.
Vehicle 2 arrived at minute 9 and began refueling with Dynamic Diesel at Pump Island 1.
Vehicle 3 arrived at minute 15 and began refueling with Primax95 at Pump Island 2.
Vehicle 2 finished refueling and departed at minute 19.
Vehicle 3 finished refueling and departed at minute 19.
Vehicle 4 arrived at minute 19 and began refueling with Primax95 at Pump Island 4.
Vehicle 4 finished refueling and departed at minute 23.
Vehicle 5 arrived at minute 25 and began refueling with Dynamic Diesel at Pump Island 3.
Vehicle 6 arrived at minute 28 and began refueling with Primax95 at Pump Island 1.
Vehicle 5 finished refueling and departed at minute 31.
Vehicle 7 arrived at minute 33 and began refueling with Primax97 at Pump Island 2.
Vehicle 6 finished refueling and departed at minute 36.
Vehicle 8 arrived at minute 39 and began refueling with Primax97 at Pump Island 4.
Vehicle 7 finished refueling and departed at minute 41.
Vehicle 8 finished refueling and departed at minute 43.
Vehicle 9 arrived at minute 45 and began refueling with Primax95 at Pump Island 3.
Vehicle 10 arrived at minute 49 and began refueling with Primax95 at Pump Island 1.

4.6 Comparison of Peak & Non-Peak Hour

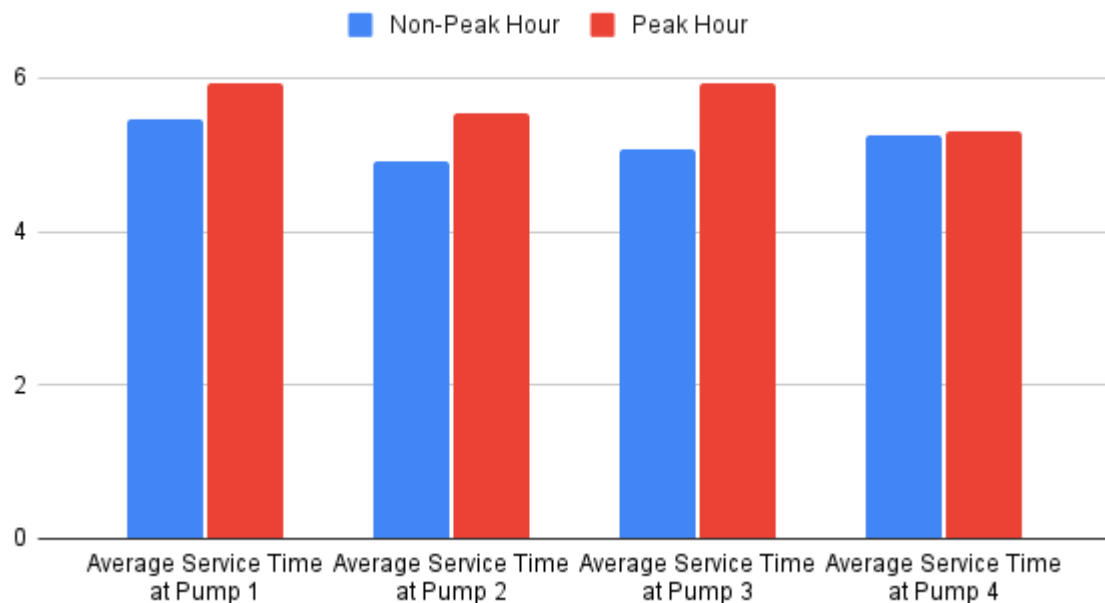
4.6.1 Column Chart of Average Waiting, Average Time in System & Probability of Waiting

Non-Peak Hour and Peak Hour



4.6.2 Column Chart of Average Service Time at Pump 1 - 4

Non-Peak Hour and Peak Hour



5.0 Analysis

5.1 Comparative Analysis of Non-Peak vs Peak Hour Performance

- **Waiting Time**
 - During non-peak hours, the average waiting time experienced by vehicles was recorded at **0 min 0 sec**. In stark contrast, under peak hour conditions, this metric surged to **0 min 58 sec**. The tighter inter-arrival time distribution during peak hours (1-2 minutes) as opposed to non-peak hours (3-6 minutes) is directly responsible for this significant rise. As cars compete for available pumps during peak hours, the more frequently arriving vehicles invariably result in a longer queue.
- **Time Spent in System**
 - A similar pattern was also seen in the average amount of time a car spent inside the petrol station system overall (from arrival to exit). Non-peak hours exhibited an average system time of **5 min 11 sec**, which escalated to **6 min 37 sec** during peak hours. This statistic highlights the entire inefficiency imposed by increased demand by fully capturing the cumulative impact of both waiting time and the actual refuelling duration.
- **Probability of Waiting**
 - There was a significant difference between the two scenarios in the probability of a vehicle experiencing a delay before service. For non-peak operations, the probability of waiting stood at **0.00**. This figure dramatically increased to **0.44** during peak hours. This noticeable change demonstrates that a car is far more likely to enter a queue at times of high demand, pointing to a larger degree of systemic congestion.

5.2 Pump Utilization & Service Times

The average servicing time at each of the four pumps was monitored by the simulation.

- **Individual Pump Performance:**

The average service times at individual pumps **Pump 1: 5.46 mins, Pump 2: 4.92 mins, Pump 3: 5.08 mins, Pump 4: 5.25 mins, for non-peak; Pump 1: 5.92 mins, Pump 2: 5.54 mins, Pump 3: 5.92 mins, Pump 4: 5.31 mins, for peak hours** showed relative consistency within each scenario. This implies that the main cause of longer wait times during peak hours is not the effectiveness of the individual refuelling procedure itself (when a vehicle reaches a pump). Rather, the speed at which cars can reach a pump that is available is the bottleneck.

- **Lane Assignment Strategy:**

In order to distribute the load, the simulation uses a random tie-breaker to allocate vehicles to one of two lanes (Lane 1 for Pumps 1 & 2, Lane 2 for Pumps 3 & 4). Regardless of the balancing mechanism, the sheer volume of arrivals during peak hours overwhelms the fixed number of service channels, even while this strategy aids in balancing utilisation. When demand is high, the system is essentially capacity-limited.

5.3 Impact of Simulation Parameters & Randomness

A realistic level of unpredictability is introduced by using random number generation for vehicle type (which affects refuelling time), petrol type selection, and inter-arrival times. Because client arrivals and servicing needs are uncertain, the simulation's stochastic nature guarantees that it offers a probabilistic depiction of petrol station operations. The inter-arrival time distribution's crucial position as the primary driver of system performance in this model is demonstrated by the notable variances between non-peak and peak hours that are a direct result of the programmed changes in this time distribution.

6.0 Conclusion

This report details a simulation study of a petrol station's operational performance, explicitly differentiating between non-peak and peak hour conditions. The findings offer valuable insights into the dynamics of waiting times, system throughput, and resource utilization under varying traffic demands.

6.1 Summary of Key Findings

The simulation clearly shows that the most important factor influencing a petrol station's effectiveness and patron experience is the rate at which cars arrive. The station has a large rise in average customer waiting times and a higher likelihood of vehicles having to queue during peak hours, which are defined by noticeably shorter inter-arrival durations. On the other hand, the station has far shorter wait times and more efficient traffic movement during off-peak hours. The restricted number of pumps (four in this form) becomes a glaring bottleneck during times of high demand, causing congestion mostly at the point of entry into the service area, even though individual pump servicing times stay largely constant.

6.2 Implications & Recommendations

The findings demonstrate the difficulties in maintaining a fixed-capacity service infrastructure when demand varies. Customer satisfaction is directly impacted by longer wait times during peak hours, and businesses may lose out if customers decide to use other services. The following suggestions are put out in an effort to lessen these problems:

- **Capacity Expansion:** Increasing service capacity is the most straightforward way to lessen congestion during peak hours. This can entail expanding the number of fuelling pumps or repurposing existing space to support more refuelling activities at once.
- **Dynamic Resource Allocation:** Put plans in place to distribute resources in real time according to anticipated demand. In order to reduce processing time per vehicle, this may entail opening all available pumps during peak hours (if some are normally closed) or even hiring more workers to speed up the payment process or manage traffic flow.
- **Optimise Queue Management:** Look into sophisticated queueing techniques, like a single, integrated queue that supplies all of the available pumps. By reducing the possibility of one pump being idle while another has a huge backlog, this can result in more effective use of service channels.
- **Demand Management:** To more fairly distribute vehicle arrivals throughout the day, long-term strategies should take into account demand management tactics including providing rewards for refuelling during off-peak hours (such as loyalty discounts).

6.3 Limitations & Future Works

Although this simulation offers insightful information, it is predicated on specific premises and has built-in restrictions that point to potential directions for further study:

- **Customer Behaviour:** Complex customer behaviours like customer abandonment—when a vehicle leaves the queue if the wait time is too long—and balking—when a vehicle decides not to join a queue at all because it looks too long when it arrives—are not taken into consideration by the current model. A more accurate depiction of lost income and consumer discontent would result from incorporating these behaviours.
- **Dynamic Distributions:** For any situation, the inter-arrival and service time distributions remain constant. To improve the model's adaptability and predictiveness, future research might include dynamically shifting distributions throughout the day (such as hourly fluctuations) or integrate real-world data.
- **System Complexity:** Only the fuelling process is the subject of the model. To replicate a more comprehensive petrol station operation and find more bottlenecks, future improvements could extend to incorporate more service components, including convenience shop purchases or vehicle wash services.
- **Validation and Sensitivity Analysis:** Several simulation runs using various random seeds would be advantageous for a more reliable analysis in order to evaluate the variability of the outcomes. Additionally, a sensitivity analysis might be carried out to determine the effects of modifications to important factors (such as the quantity of pumps or the average refuelling times) on the overall performance of the system.
- **Visual Simulation:** By creating a visual animation or graphical user interface (GUI), the simulation's interpretability would be greatly improved, enabling a more natural comprehension of vehicle flow, queue dynamics, and pump utilisation over time.

7.0 Flowchart

